



ASSIGNMENT

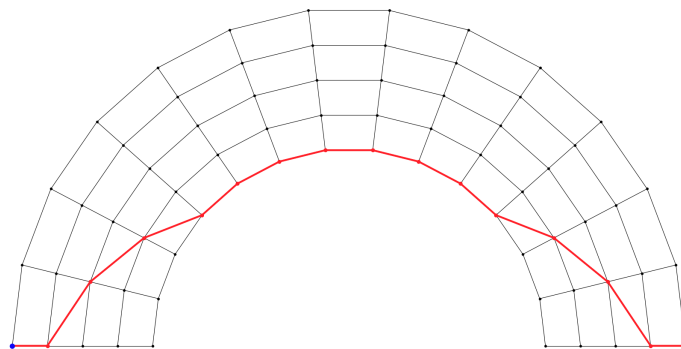
algorithmic

The SGe-09 driverless car must travel along a steep 180° turn, and you must help it plan its path!

The track has an inner radius of $I=7\text{m}$, an outer radius of $O=10\text{m}$, and spans a semicircle of $T=180^\circ$. You decide to split the track in discrete nodes aligned in a circular grid of $K=128$ points in the direction of travel by $H=24$ points in the radial direction. Movements of the car are modeled as travelling from one point of the grid to an adjacent one, in either direction or by going diagonally; each point not on the border is thus adjacent to 8 other points. Distance between two points is defined in the usual euclidian sense.

The car travels counterclockwise, must start from the outmost (furthest from the center) point on the right and reach the outmost point on the left. You are tasked with finding a shortest path on the grid. Output the length of a shortest path.

Code the problem in any language (preferred: C++ or Python) and submit your public git repository.



Example of a shortest path on a smaller ($K=14$, $H=5$) circular grid.